

JYOTHIRADITHYA MADAPATI

Aspiring SOC Analyst | Cybersecurity Undergraduate, IIIT Kottayam

madapatijyothiradithyamadapati@gmail.com | +91 99491 53105 | [LinkedIn](#) | [GitHub](#)

SUMMARY

Cybersecurity undergraduate with hands-on experience building a virtualized SOC lab, deploying SIEM (Wazuh) for log monitoring, and developing a Python-based vulnerability scanner that identified a real high-severity CVE. Completed job simulations with Deloitte and TCS focused on cybersecurity analyst workflows. Seeking a SOC Analyst / Blue Team internship to apply threat detection and network monitoring skills.

EDUCATION

Indian Institute of Information Technology, Kottayam (IIIT Kottayam)

Expected 2028

B.Tech in Computer Science & Engineering (Cybersecurity Specialization) — 3rd Year

TECHNICAL SKILLS

- Security Tools: Wazuh, Nmap, Wireshark, VirtualBox
- Networking: Network Security, Network Traffic Analysis, DNS, TCP/IP, Secure Shell (SSH)
- Security Concepts: Vulnerability Assessment, Threat & Vulnerability Management, Information Security Awareness
- Operating Systems: Kali Linux, Ubuntu, Linux, Virtual Machines (VM)
- Programming: Python
- Other: Problem Solving, Project Management, Communication

PROJECTS

VulnProbe — Python Network Vulnerability Scanner

Jun 2026

- Built a Python tool automating core vulnerability-assessment steps: port scanning (sockets), service/version fingerprinting (Nmap integration), and CVE cross-referencing via the NVD REST API.
- Auto-generated color-coded HTML vulnerability reports; during testing, identified a real HIGH-severity CVE (CVE-2021-44224) in an outdated Apache httpd service.

GitHub: github.com/Adithya9x/VulnProbe

Cybersecurity Home Lab & SOC Monitoring

Jun 2026

- Designed and built a virtualized SOC lab (Kali Linux + Ubuntu Server/Desktop in VirtualBox) with host-only networking and secure OpenSSH remote access between VMs.
- Deployed Wazuh SIEM to monitor authentication logs and failed login attempts; simulated an SSH brute-force attack (Kali → Ubuntu) to validate detection and logging workflows.
- Diagnosed and resolved real infrastructure issues (resource exhaustion, CPU soft lockups) by managing system resources and disabling non-essential services.

Network Traffic Analysis using Wireshark

Jun 2026

- Captured and analyzed live Wi-Fi traffic (DNS, HTTP, HTTPS/TLS, TCP); correlated DNS query/response pairs via transaction IDs and inspected TLS Client Hello packets to extract SNI metadata.
- Performed anomaly detection using TCP resets, retransmissions, timing patterns, and port usage; documented findings in a SOC-style incident report.

GitHub: github.com/Adithya9x/network-traffic-analysis-wireshark

PhantomGrid — Behavioral Biometrics Engine (Team Project, PSBs Hackathon 2026)

- Built the frontend behavioral signal capture layer (capture.js), tracking user interaction patterns (tap decoy timing, dwell time, key entry timing) during transaction flows.
- Packaged captured signals into a behavioral JSON payload sent to the backend via POST /enroll and POST /verify endpoints, supporting a 3-layer biometric verification system.

Sentinel — Deception-Based SOC Detection Platform

Jul 2026

- Designed a honeypot-based detection system that eliminates false positives by architecture: fake AWS credentials, SSH keys, and .env files are planted so any interaction with them is a confirmed attacker action, not a probabilistic alert.
- Built a Python file-monitoring engine (watchdog/inotify) that detects honeypot access in real time, tags each event with the relevant MITRE ATT&CK technique, and triggers desktop notifications.
- Developed a live Flask dashboard (Live Alerts, Honeypot Inventory, MITRE Coverage) that auto-launches the moment a honeypot fires.
- Validated the system end-to-end against a real simulated attacker (Kali Linux over SSH) accessing honeypots on an Ubuntu victim host.

GitHub: github.com/Adithya9x/sentinel-deception

CERTIFICATIONS & JOB SIMULATIONS

- Cybersecurity Fundamentals — IBM
- Cybersecurity Analyst Job Simulation — Tata Consultancy Services (via Forage)

- Cyber Job Simulation — Deloitte (via Forage)
- Career Essentials in Cybersecurity — Microsoft & LinkedIn Learning (Threat & Vulnerability Management, Information Security Awareness)
- Introduction to Cybersecurity — Cisco Networking Academy